P.8 Sequences; Summation Notation; the Binomial Theorem

Name _____ Period ____ Date _____

AP Calculus P.8 · Student sheet

I can…

- write the terms of a sequence, including a recursive sequence.
- use summation notation and find sums.
- evaluate binomial coefficients and expand binomials with the Binomial Theorem.

Vocabulary · fill in the missing word

Sequence — a _____ whose domain is the set of positive integers

Recursive sequence — a sequence given by its first _____(s) and a rule for each term from the ones before it

Factorial — n! = _____ for n ≥ 1, and 0! = 1

Binomial coefficient — C(n, j) = _____

Part 1 · Pascal's pattern

Build Pascal's triangle (row 0: 1; row 1: 1 1), then expand.

Row 2	Row 3	Row 4	Row 5

$(x + 1)^2$ = _____ $(x + 1)^3$ = _____

Part 2 · Notes and examples (fill in each note, then work the example)

Sequences: $\{s_n\}$ lists $s_1, s_2, s_3, \ldots$; the domain is the _____ integers.

Example 1

Write the first five terms.

a. $a_n = \dfrac{n}{n+1}$

b. $b_n = (-1)^n n^2$

Example 2

Write the first five terms.

a. $a_1 = 2$, $a_n = n + a_{n-1}$

b. $a_1 = 1$, $a_n = 3a_{n-1}$

Summation properties: $\Sigma\, c = nc$, $\Sigma\, k = n(n+1)/2$, $\Sigma\, k^2$ = _____ (each sum from k = 1 to n).

Example 3

Find each sum.

a. $\Sigma\,(2k - 1)$, $k = 1$ to 100

b. $\Sigma\,(4k + 3)$, $k = 1$ to 18

Binomial coefficient: $C(n, j) =$ _____

Example 4

Evaluate $C(5, 2)$, $C(8, 4)$, and $C(100, 2)$.

Binomial Theorem: $(x + a)^n = \Sigma\, C(n, j)\, a^j\, x^{n-j}$, $j = 0$ to n: the coefficients are row n of _____ triangle.

Example 5

Expand $(x + 2)^4$.

Part 3 · Partner practice

Partner 1

Find $\Sigma\ 6$ ($k = 1$ to 50) and $\Sigma\ 3k$ ($k = 1$ to 30).

Partner 2

Expand $(2x - 1)^3$.